

Towards an Immune System for Large Language Models: The Five-Anchor Audit Framework and Four-Tier Detection Chain

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Abstract

As large language models (LLMs) become embedded in critical decision-making pipelines, the threat of coordinated data poisoning has escalated from a theoretical concern to an active attack surface. Current defenses—RLHF, content filtering, adversarial training—are passive, reactive, and fundamentally un-auditable. We propose a paradigm shift: an *immune system* for AI that detects, contains, and learns from poisoned data through a multi-layered framework consisting of (1) a Four-Tier Detection Chain that escalates from publisher credibility to cross-cluster gang identification, (2) a Five-Anchor Audit system that prevents single-point verification failures, and (3) a Three-Line Defense architecture that closes the loop from metadata to physical reproducibility. Unlike prior work that treats data poisoning as input sanitization, our framework treats it as an immune challenge—the system learns from each exposure, building calibration memory over time. We open-source the detection methodology and audit standards to enable community validation and iterative hardening.

1 Introduction

The volume of AI-generated content is growing exponentially. So is the volume of AI-generated *poison*. Coordinated data poisoning attacks are no longer carried out by individual bad actors inserting single artifacts; they are now industrialized operations producing self-referential networks of fabricated papers, datasets, and citation chains that appear internally consistent but are systematically detached from empirical reality [1].

Current approaches to AI safety fall into three categories: reinforcement learning from human feedback (RLHF), content filtering pipelines, and adversarial training. All three share a structural weakness: they operate on the *output* side of the model, after the training data has already been ingested. They are passive filters, not active immune systems. They cannot answer the question: *how do we know this training data point was real?*

We argue that what is needed is not better filtering, but a fundamentally different architecture—an **immune system** that:

1. Detects poisoned data *before* it enters the training pipeline
2. Provides auditable evidence chains for every detection decision
3. Learns from each exposure to build calibration memory
4. Allows false positives to be challenged and corrected

5. Closes the loop through physical reproducibility verification

This paper presents the design, methodology, and initial calibration standards for such a system.

2 Conceptual Architecture: From Layered Defense to Cognitive Sphere

Before presenting the specific detection and audit mechanisms, we establish the conceptual architecture that governs their interaction. Traditional AI safety frameworks adopt a layered (defense-in-depth) model, where detection, auditing, logging, and response are arranged hierarchically. We argue that the layered model carries an implicit architectural bias: it assigns priority ordering to dimensions that should function as mutually calibrating equals. A layer that sits “above” is implicitly granted veto power over layers “below”—a structural vulnerability when the higher layer itself can be compromised.

2.1 The Cognitive Sphere Model

We propose an alternative: the **Cognitive Sphere**, in which six core dimensions—Detection, Detoxification, Audit, Evidence Preservation, Historical Memory, and Signal Stitching—are not arranged in layers but distributed as equidistant points on a spherical surface. Each dimension operates concurrently with equal architectural standing; no dimension is structurally subordinate to any other.

The six dimensions are:

1. **Detection:** Identifying potential poison through the four-tier chain.
2. **Detoxification:** Removing or quarantining confirmed poison from training pipelines.
3. **Audit:** Verifying that detection and detoxification decisions are well-founded and free from systemic bias.
4. **Evidence Preservation:** Immutable logging of every decision, its evidentiary basis, and its disposition.
5. **Historical Memory:** Maintaining calibrated priors across exposure events so that the system learns from past poison patterns.
6. **Signal Stitching:** Connecting weak signals across dimensions and temporal windows to identify emergent threats that no single dimension would detect alone.

The sphere model eliminates the implicit hierarchy of layered architectures. In a layered model, a compromised audit layer can silence the detection layer below it. In the sphere model, every dimension can raise an alert independently, and alerts are cross-calibrated rather than vertically adjudicated. The practical consequence is that no single compromised dimension can suppress detection signals from other dimensions.

2.2 Observation-Contemplation Duality

Each dimension within the sphere operates through two complementary modes: Observation and Contemplation. These are not binary opposites but two facets of a single cognitive act, analogous to the structural insight from Chinese classical philosophy where the hexagrams *Guan* (observing/scanning upward) and *Lin* (approaching/judging downward) together describe complete cognition.

In our framework:

- **Observation:** A passive, exhaustive scanning mode. Every data point is seen without pre-filtering. The observer does not interpret—it records. This is the “look up and survey” posture: systematic, panoramic, leaving no blind spots.
- **Contemplation:** An active, judgment-bearing mode. Collected signals are weighed, compared against historical patterns, and assigned confidence. This is the “look down and bear witness” posture: reflective, evaluative, producing actionable verdicts.

In operational terms, the Observation–Contemplation duality manifests in every anchor of the Five-Anchor Audit Framework (Section 5). The Source Anchor requires Observation (tracing provenance exhaustively) and Contemplation (judging whether the provenance chain is trustworthy). The Cross Anchor requires Observation (gathering multiple independent sources) and Contemplation (determining whether they genuinely corroborate). Neither mode alone suffices: Observation without Contemplation produces unprocessed signal noise; Contemplation without Observation produces confident judgments detached from evidence.

The ratio of Observation to Contemplation intensity—the **OC ratio**—is a configurable parameter within each dimension. A detection dimension may operate at a high OC ratio (intensive scanning, conservative judgment), while an evidence preservation dimension may operate at a low OC ratio (recording is automatic; judgment is deferred). The OC ratio is not fixed; it is dynamically adjusted based on threat posture and historical calibration feedback.

2.3 Curvature of Integrity

We introduce the concept of **Integrity-Guardianship** (Shouzheng): a depersonalized, decentralized, auditable systemic force that maintains the health of the entire sphere. Integrity-Guardianship is not a property of any individual actor—it does not depend on the benevolence of a reviewer, the good faith of an auditor, or the diligence of a detox engineer. It is a property of the system’s architecture, encoded in its code, its open-source verifiability, its on-chain audit logs, and the multi-stakeholder checks that prevent any single party from unilaterally altering verdicts.

The **Curvature of Integrity** provides a unified metric for system health. Conceptually, it describes the “tension” in the sphere: whether the six dimensions are exerting balanced calibrating force on one another. Two pathological states exist:

- **Hyper-curvature (over-defensive):** Every dimension is set to maximum sensitivity. False positive rates spike. Legitimate content is quarantined. The sphere collapses inward, suffocating the training pipeline.
- **Hypo-curvature (under-defensive):** Dimensions are too loosely coupled. Detection signals fail to propagate. Poison passes through. The sphere flattens, offering no resistance.

Optimal curvature is achieved when each dimension exerts sufficient calibrating force to sustain mutual accountability without triggering cascading false positives. We further define **local**

curvature for individual dimensions: a detection dimension with high OC ratio (scan-heavy) contributes differently to global curvature than an evidence preservation dimension with low OC ratio (record-heavy). The global curvature of the sphere is the aggregate of these local contributions, representing the overall tension state of the system. Monitoring global curvature over time provides a leading indicator of emerging defense pathologies before they manifest in detection failures.

2.4 Dao-Fa Integration: Principles Governing Technical Implementation

Every technical architecture rests on philosophical foundations, whether acknowledged or not. We make these foundations explicit through the **Dao-Fa Integration Layer**, which governs the relationship between architectural principles (Dao) and operational rules (Fa).

- **Dao (Architectural Principles):** The design philosophy that ensures the system’s direction does not deviate. In our framework, Dao manifests as four fixed anchors—Kantian universalizability (every detection rule must be defensible as a universal principle), Popperian falsifiability (every verdict must specify conditions under which it would be overturned), Deweyan pragmatic inquiry (the system must learn and adapt from experience rather than enforce static dogma), and Wang Yangming’s unity of knowledge and action (detection without remediation is incomplete knowledge).
- **Fa (Operational Rules):** The concrete scoring rubrics, pass/fail thresholds, escalation procedures, and remediation protocols described in Sections 4–8. Fa is the executable specification.

The Dao-Fa relationship is not a binary choice between philosophy and engineering, but a unified framework: Dao without Fa is direction without execution (a compass without a ship); Fa without Dao is execution without direction (a ship without a compass). The four fixed anchors provide the Dao layer with concrete, operational substance:

- **Kantian anchor:** All detection standards must be universalizable—if a rule cannot be defensibly applied to all content regardless of source, it is discriminatory architecture, not detection.
- **Popperian anchor:** Every poisoning verdict must specify the falsification condition—what evidence would reverse the judgment. This prevents unfalsifiable accusations.
- **Deweyan anchor:** Calibration is continuous. The framework does not claim to have achieved perfect detection; it claims to have established a process for asymptotic improvement.
- **Wang Yangming anchor:** Knowledge and action are unified. Detecting poison without acting to remove it, or removing poison without documenting the rationale, is a structural failure equivalent to not detecting it at all.

These anchors are not abstract philosophical decoration. They serve as the *constitutional foundation* against which every operational rule (Fa) is validated: if a proposed detection rule violates any of the four anchors, it is rejected at the design stage. This Dao-Fa integration ensures that the system’s technical evolution remains aligned with its foundational principles, preventing the gradual drift toward either over-defensive hyper-curvature or under-defensive hypo-curvature.

3 Threat Taxonomy: The Poison Data Landscape

We classify data poisoning attacks into four structural types, ordered by detection difficulty:

3.1 Type I: Publisher-Level Poisoning

Fabricated papers published through predatory journals, hijacked conference proceedings, cloned journal titles, and fake ISSN registrations. These are the most common and easiest to detect—the publishing channel itself provides strong signals.

3.2 Type II: Author-Level Poisoning

Papers from fabricated authors, identity-theft cases (real institution, fake role), impossible publication timelines (35 papers/year), and closed co-author circles where a fixed group of authors only cite and co-publish with each other.

3.3 Type III: Cluster-Level Poisoning

The most sophisticated and dangerous form. Multiple papers form a self-referential citation network across different publishers and journals. Individually, each paper may appear legitimate. Collectively, they form a closed validation loop with no external verification. This is the “logically perfect, empirically empty” attack pattern.

3.4 Type IV: Lone Perfect Poison

A single paper from a genuine author in a legitimate journal, with internally consistent methodology, but privately fabricated experimental data. No cluster exists to detect. This type bypasses publisher, author, and cluster checks—requiring physical reproducibility verification as the final defense.

3.5 Type V: Logical Fallacy Embedding

Poisoned reasoning chains that embed formal fallacies (affirming the consequent, circular reasoning), informal fallacies (appeal to authority, false dichotomy), and quantitative fallacies (Simpson’s paradox, survivorship bias) within otherwise well-structured arguments.

Table 1: Detection coverage by poison type and defense layer

Poison Type	Tier I (Publisher)	Tier II (Author)	Tier III (Cluster)
Predatory journal	✓	-	-
Fabricated authors	-	✓	-
Citation cartels	-	-	✓
Lone perfect poison	-	-	-

4 The Four-Tier Detection Chain

4.1 Design Principles

The detection chain follows four architectural principles:

- **Open reproducibility:** Any party running the same standards should reach the same conclusion
- **Full traceability:** Every score can be traced back to specific evidence
- **Multi-dimensional cross-validation:** Failure of a single dimension does not cause systemic misjudgment
- **No single point of failure:** The scoring system does not depend on the absolute correctness of any single component

4.2 Tier I: Publisher Credibility Scoring

Weight: 20% of composite score.

Five sub-dimensions, each scored 0–20:

1. **Index inclusion:** SCI/SSCI/AHCI indexing status. Predatory journals: -10 ; fake ISSN: -25 .
2. **Publisher history:** Years of operation. < 3 years: -10 ; < 1 year: -15 .
3. **Peer review process:** Transparent review procedures. “Rapid publication” promises: -20 .
4. **Editorial board:** Verifiable academic backgrounds. Fake board members: -20 .
5. **Database coverage:** Major academic database inclusion. Google Scholar only: -10 .

Pass threshold: $\geq 60/100$. Below 60 triggers immediate flagging regardless of other tier scores.

4.3 Tier II: Author Profiling

Weight: 25% of composite score.

Five sub-dimensions:

1. **Identity verifiability:** ORCID, institutional directory. Unverifiable: -10 .
2. **Domain consistency:** Publication scope concentration. ≥ 3 unrelated domains: -15 (single red flag).
3. **Publication timeline:** Temporal plausibility. Anomalous productivity (> 1 paper/week sustained): -10 . Future citations: -20 .
4. **Co-author network:** Verifiable collaborators. Circular co-author loops: -20 .
5. **Academic trajectory:** Career progression coherence. Sudden domain jumps without transition: -15 .

Pass threshold: $\geq 60/100$.

4.4 Tier III: Cluster Detection

Weight: 30% of composite score.

This is the most critical detection layer. While Tiers I–II examine individual papers, Tier III examines *relationships* between papers:

1. **Publisher clustering:** ≥ 3 related papers from the same publisher: -10 .
2. **Author circle clustering:** High co-authorship/citation overlap within a small group: -15 .
3. **Temporal clustering:** Concentrated publication windows: -15 .
4. **Citation loops:** Mutual citation forming closed loops: -20 ; multi-level loops: -25 .
5. **Content fingerprinting:** Highly consistent writing patterns across supposedly different authors: -15 .

Key insight: A single paper with mediocre Tier I–II scores may be merely low-quality. Multiple papers forming a detection cluster is strong evidence of deliberate fabrication—regardless of individual tier scores.

4.5 Tier IV: Cross-Cluster Gang Identification

Weight: 25% of composite score.

Tier IV operates across previously detected clusters to identify coordinated groups:

1. **Cross-domain author mapping:** Overlapping authors across domain-separated clusters: -20 .
2. **Content synergy:** Papers that mutually “prove” each other’s claims: -25 ; synergy chains ≥ 3 papers: -30 .
3. **Temporal orchestration:** Staggered publication timing suggesting scripted sequencing: -15 .
4. **Funding source convergence:** Multiple papers acknowledging the same non-existent funding body: -10 .
5. **Infrastructure sharing:** Same GitHub org, same submission server, identical acknowledgment templates: -10 .

4.6 Composite Verdict Matrix

Table 2: Composite verdict matrix across four tiers

Tier I	Tier II	Tier III	Tier IV	Verdict
≥ 70	≥ 70	≥ 60	≥ 60	Trusted
60–69	60–69	40–59	40–59	Suspect
< 60	< 60	< 40	< 40	Poison
any	any	any	≥ 2 hits	Gang Poison

5 The Five-Anchor Audit Framework

Single-source verification is structurally unreliable. Any single data point, no matter how credible its source, may be wrong. The five-anchor framework ensures that every piece of information is cross-validated across five independent dimensions:

1. **Source Anchor** (): Where does this information come from? What is the chain of custody? Analogous to canonicalization in religious traditions—only texts with verified provenance enter the trusted corpus.
2. **Logic Anchor** (): Is the reasoning chain internally consistent? Does it survive adversarial examination? Analogous to the scholastic tradition of disputation, where every proposition must withstand counter-arguments.
3. **Interest Anchor** (): Who benefits from this information being accepted as true? Are funding sources, institutional affiliations, and conflicts of interest fully declared and independently verified?
4. **Compliance Anchor** (): Does the information comply with established standards? Compliance is not binary—standards evolve from recommendation to requirement based on conditional triggers (scale, external threat, trust crisis, conflict of interest).
5. **Cross Anchor** (): Can the same conclusion be reached through multiple independent sources? This anchor is explicitly designed to prevent the single-tribunal problem—history demonstrates that single-authority judgments of “heresy” are systematically unreliable.

The five anchors are not additive—they are multiplicative. One anchor failing to verify is sufficient to flag the information for review, regardless of the other four. Within each anchor, the Observation–Contemplation duality (Section 2) operates concretely: the Source Anchor’s Observation mode traces provenance chains exhaustively, while its Contemplation mode judges whether those chains are trustworthy; the Cross Anchor’s Observation mode gathers independent sources, while its Contemplation mode determines genuine corroboration versus coordinated mimicry.

6 Information Completeness: A Root Cause of Attribution Hallucination

The detection and audit mechanisms described in Sections 3–5 address threats originating from *external* data manipulation. However, a second class of threat operates through a different mechanism: the model’s own inference engine generating plausible but causally incorrect attributions when critical information is missing. This section formalizes the relationship between information gaps and attribution hallucination, presents a dual-perspective correction methodology, and integrates the resulting audit capability into the Five-Anchor framework.

6.1 Problem Definition: The Info-Gap Hypothesis

We define **attribution hallucination** as the phenomenon where a model generates a causal explanation that appears internally consistent but contains at least one link unsupported by verifiable evidence. Our central hypothesis is:

$$H_{\text{attribution}} \iff \text{info-gap} \wedge \text{strong-inference-closure} \quad (1)$$

That is, attribution hallucination occurs exactly when two conditions are simultaneously satisfied: (1) an information gap exists—the model lacks access to facts necessary for sound causal attribution, and (2) the model possesses a strong inference engine that automatically fills the gap with “plausible” causal links derived from statistical patterns rather than empirical evidence.

Without Gap	With Gap
$\text{Event}_A \xrightarrow{\text{Verified}} \text{Event}_B$	$\text{Event}_A \xrightarrow{\text{Inferred}} \text{Motive}_X \xrightarrow{\text{Inferred}} \text{Event}_B$
All causal links independently verifiable	At least one link is a statistical bridge
Low hallucination risk	High hallucination risk

Figure 1: Info-gap as the structural precondition for attribution hallucination. The right column shows an inferred motive inserted to bridge a causal gap.

6.1.1 Causal Link Taxonomy

For audit purposes, we classify every link in a causal attribution chain into three types:

Type	Name	Definition
A	Directly Verifiable	Both endpoints independently confirmable via external sources
B	Partially Verifiable	One endpoint independently confirmable; the other inferred
C	Unverifiable Bridge	Neither endpoint independently confirmable; purely statistical

Table 3: Causal link classification for attribution audit.

Trigger Rule: Any Type C link in a final attribution triggers `d2.5_c_gap_trigger`, reducing attribution confidence proportionally to the ratio of Type C links in the total causal chain.

6.1.2 A Counter-Intuitive Prediction

A distinguishing prediction of this framework is that **stronger models produce more attribution hallucination in info-gap scenarios** (see §6.6 for experimental design and predictions):

1. Stronger model $\Rightarrow$ stronger inference engine $\Rightarrow$ faster inference-closure
2. With the info-gap held constant, faster closure $\Rightarrow$ more confident but less detectable hallucination
3. Weaker models are naturally safer in this dimension: insufficient inference capability prevents gap-filling

This runs counter to the common assumption that model scaling universally improves reliability, and provides a falsifiable prediction for empirical testing.

6.2 Methodology: Dual-Perspective Attribution Correction

We present a five-stage protocol that combines two complementary perspectives: *progressive gap exposure* (identifying where inference has substituted for evidence) and *incentive traceback* (tracing stakeholder incentives to verify or refute inferred causal links).

6.2.1 Stage 1: Progressive Gap Exposure

Strong inference engines produce “closed” causal narratives where inferential jumps are indistinguishable from evidence-based links. The solution is not to suppress inference, but to systematically expose the gaps through multi-turn open-ended probing:

1. For a given attribution claim C , identify all unverified premises $\{p_1, p_2, \dots, p_n\}$ that C depends on
2. For each premise p_i , ask: “What evidence would falsify this premise?”
3. If no falsification condition can be specified, mark the link as Type C
4. Iterate: each exposed gap may reveal sub-gaps at finer granularity
5. Terminate when the gap set is empty or all remaining gaps are unqueryable

This protocol is analogous to counterexample search in interactive proof systems, adapted for the non-automated context of human-AI collaborative attribution.

6.2.2 Stage 2: Information Source Completion

Once gaps are identified, the model performs targeted external retrieval for each gap. Unlike generic RAG, which retrieves broadly, this step retrieves specifically to fill identified causal gaps:

- For each gap g_i , issue a directed query to external knowledge sources
- Verify that retrieved information directly addresses the identified gap
- Re-classify links after retrieval: Type C $\rightarrow$ Type B or Type A

The principle: hallucination grows in information voids. Models fill empty spaces automatically, and the fill material is “plausibility” rather than “truth.” The solution is not to suppress inference, but to actively complete the information landscape—once the gaps are filled with evidence, inference no longer needs to bridge them.

6.2.3 Stage 3: Incentive Chain Traceback

For attribution claims involving multiple stakeholders, we trace the incentive structure of each stakeholder. The technical approach formalizes two co-isomorphic priors:

Short-Chain Prior (Ockham) Shorter causal explanations are preferred over longer ones, all else being equal.

Internal-Incentive Prior Internal incentive explanations are preferred over external coercion explanations, all else being equal.

These two priors are structurally isomorphic: both prefer explanations with fewer unobserved intermediaries. The combined rule—the **Self-Interest Prior**—states: *exhaust internal incentive explanations before invoking external intervention.*

6.2.4 Stage 4: Self-Interest Prior Convergence

Given the retrieved information sources S and the stakeholder incentive map I , the correction algorithm converges on the most parsimonious explanation:

1. Generate the shortest causal chain consistent with all verified sources S
2. Identify the most sufficient internal incentive explanation from I
3. If the internal incentive explanation is sufficient, adopt it
4. Otherwise, adopt the shortest-chain explanation
5. If neither is sufficient, mark the attribution as UNCERTAIN

6.2.5 Stage 5: Attribution Verification

After correction, verify that the revised attribution chain contains no unacknowledged inference bridges. If any Type C links remain:

- Mark them explicitly with [INFERRED] tags
- Compute confidence as $1 - (n_C/n_{\text{total}})$, where n_C is the number of Type C links
- Flag for human review if confidence falls below threshold

6.2.6 Algorithm

```
function dual_perspective_correction(claim, domain):
    // Stage 1: Progressive Gap Exposure
    gaps = open_ended_probe(claim)
    if gaps.empty() or gaps.all_unqueryable():
        return {verdict: GAP_BLOCKED}

    // Stage 2: Information Source Completion
    sources = [external_retrieval(g) for g in gaps]

    // Stage 3: Incentive Chain Traceback
    stakeholders = identify_stakeholders(domain)
    incentive_map = {s: compute_incentive(s, claim) for s in stakeholders}

    // Stage 4: Self-Interest Prior Convergence
    // co-isomorphic: both priors select the most parsimonious explanation
    short_chain = shortest_causal_explanation(sources)
    internal = most_sufficient_internal_incentive(incentive_map)

    correction = internal if internal.sufficient else short_chain

    // Stage 5: Verification
    if correction == null or correction.depends_on("inference_bridge"):
        inference_bridges = list_inference_bridges(claim)
        return {verdict: UNCERTAIN,
                gaps: gaps,
                gap_types: classify_gaps(gaps),
```

```

inference_bridges: inference_bridges,
confidence: len(inference_bridges) / total_links(claim)}

return {verdict: CORRECTED, correction: correction}

```

6.3 Degradation Spectrum

The methodology’s effectiveness degrades along a spectrum determined by the nature of the information gap:

Stage	Description
Fully Queryable	Complete correction possible; all gaps can be exposed and filled
Partially Queryable	Partial correction: gaps can be identified but not all filled; remaining uncertainty is quantified
Unqueryable	Degraded: can only flag global uncertainty without offering alternative attribution
Hidden Gap	Worst case: the info-gap itself is masked by inference closure; methodology fails silently

The `GAP_BLOCKED` return in the algorithm covers the latter two stages. The first two stages are differentiated through the `UNCERTAIN` return’s quantified confidence value and gap-type classification.

6.4 Comparison with Existing Approaches

Approach	Mechanism	Limitation
RLHF/RLAIF	Preference labeling suppresses undesired outputs	Behavioral constraint; toms, not causes
RAG	Retrieval fills knowledge gaps	Fills factual gaps only; d causal attribution chains
Multi-Agent Voting	Consensus via debate or majority	Shared info-gap → share is a blind spot
Debate + Human Judge	AI debate with human arbitration	Approaches our method versarial; we prioritize correction
Hallucination Governance Pipelines (2024–2026)	Integrated detect-mitigate-govern engineering pipelines	Detection alone does not ified content from reachi audiences—the gap this
This Work	Gap exposure + incentive trace-back + Self-Interest Prior	Cognitive audit: addre rather than output beha

Table 4: Comparison with existing hallucination mitigation approaches.

While each individual component (gap detection, retrieval, incentive modeling) has academic precedent, the integration of all components into a unified cognitive audit architecture is novel. The key differentiator is treating hallucination not as an output quality problem but as a structural consequence of the inference engine operating with incomplete information.

6.5 Framework Integration

The methodology integrates into the existing Five-Anchor framework at six specific touchpoints, strengthening the audit capability without adding new anchors or disrupting existing structures:

6.5.1 D2: Logic Anchor — Attribution Audit

- **d2.4_info_gap_detection:** Add a column to the logic audit matrix classifying each causal link as Type A/B/C
- **d2.5_c_gap_trigger:** Any Type C link triggers automatic confidence downgrade for the attribution verdict

6.5.2 L5: Causal Attribution Layer — New Sub-layers

- **L5b_interest_chain:** Incentive traceback sub-module that maps stakeholder incentives for any multi-party attribution claim
- **L5c_self_interest_prior:** Self-Interest Prior convergence module that selects the most parsimonious internal-incentive explanation

The updated detection chain structure becomes:

L1: Basic Fact Matching
L2: Semantic Consistency
L3: Logical Closure
L4: Statistical Bias Detection
L5: Causal Attribution Audit
 L5a: Surface Causality Check
 L5b: Incentive Chain Traceback *[new]*
 L5c: Self-Interest Prior Convergence *[new]*
L6: Counter-Cognitive-Colonization Audit

6.5.3 D5: Methodology Anchor — New Rules

- **d5.5_prior_rule:** Self-Interest Prior as a cross-anchor convergence rule—when multiple anchors produce conflicting verdicts on attribution claims, the prior guides resolution
- **d5.6_info_threshold:** Maximum allowable Type C link ratio in any core causal chain, initially set conservatively at ≤ 1 . This parameter is flagged for dynamic adjustment via FML calibration data

Implicit Assumption Correction. The info-gap hypothesis directly challenges an unstated assumption in the methodology anchor: that sufficiently good methodology can compensate for missing information. Our analysis shows the opposite—stronger methodology accelerates gap-closure, making hallucination more confident and less detectable. Therefore, **d5.6_info_threshold** functions as a brake mechanism rather than an optimization target.

6.5.4 FML: Failure Memory Layer

Info-gap patterns are stored in the Failure Memory Layer for future pre-triggering. When a new attribution task matches a stored gap pattern, the system automatically reduces inference speed and increases verification requirements before generating attribution.

6.5.5 Tier IV: Cross-Cluster Detection

A new coroutine `tier4_info_gap_coroutine` detects whether apparent multi-source consensus actually originates from a shared information gap—a pattern where multiple sources independently confirm the same causal narrative because they share the same missing information, not because the narrative is true.

6.5.6 D6: Language-View Layer

Attribution audit tags are added to the language-view annotation schema: [GAP] for identified information gaps, [INFERRED] for inference-bridge links, and [VERIFIED] for independently confirmed causal links.

6.6 Experimental Design and Predictions

The experimental methodology follows directly from the info-gap hypothesis (§1): construct a fixed set of info-gap scenarios (missing key stakeholder motivation information), expose multiple models of varying capability (7B/13B/70B/DeepSeek-R1) to identical scenarios, and measure the ratio of Type C links in causal attributions alongside independent fact-checking accuracy. A control condition uses the same scenarios with info-gaps filled. This design isolates the effect of inference capability on attribution quality independent of scenario difficulty.

The framework generates specific, falsifiable predictions suitable for empirical validation:

1. **Model Scaling × Hallucination Rate.** In controlled info-gap scenarios, larger models will exhibit a higher Type C link ratio than smaller models, while simultaneously expressing higher confidence in their attributions. This creates a “confidence-hazard inversion” where stronger models are more dangerous for attribution tasks with incomplete information.
2. **Gap-Filling Efficacy.** Applying the five-stage protocol will reduce Type C link ratios from a baseline (estimated 20–40% for strong models in info-gap conditions) to below 5%, with the majority of remaining Type C links classified as Known Unknowns.
3. **Self-Interest Prior Accuracy.** In multi-stakeholder scenarios where ground-truth incentives are independently verifiable, the Self-Interest Prior will correctly identify the dominant causal factor in $\geq 80\%$ of cases, outperforming both pure chain-of-thought reasoning and external-intervention-first heuristics.
4. **Shared Info-Gap Detection.** The Tier IV gap-coroutine will identify shared info-gap consensus patterns with a false positive rate below 15% in controlled multi-source scenarios.

6.7 Applicability Boundaries

The methodology explicitly does not apply to all hallucination types. Its effectiveness is limited to **attribution hallucination**—cases where the model constructs causal narratives with information gaps. It is not designed for:

- Factual hallucination (missing knowledge): better addressed by retrieval augmentation and confidence calibration
- Statistical bias hallucination: requires rare-signal amplification methods

- Semantic ambiguity hallucination: requires disambiguation anchoring
- Adversarial injection: requires security-hardening mechanisms
- Value-alignment hallucination: requires broader calibration framework beyond any single methodology

Within its domain, however, the methodology addresses a failure mode that existing approaches systematically miss: the *invisible* hallucination that appears structurally sound but contains unverifiable causal links.

7 False Positive Remediation: Auditing the Audit Chain

An immune system that cannot correct its own mistakes is not an immune system—it is an inquisition. We design a four-stage remediation pipeline:

7.1 Stage 1: Automatic Hold

When Tier IV (gang identification) triggers without Tier I–III support, the verdict is automatically held pending human review. This prevents the highest-confidence (and highest-risk) judgments from executing without oversight.

7.2 Stage 2: Independent Review Tribunal

A panel of ≥ 3 reviewers, including at least 1 external expert, conducts a blind re-review. The panel does not see the original scores, preventing anchoring bias.

7.3 Stage 3: Graduated Recovery

Verdicts are not binary (poison/clean). Recovery follows a five-level gradient:

- **Confirmed Poison:** Appeal rejected, mark retained
- **Downgraded to Suspect:** Evidence insufficient for poison, but doubts remain
- **Insufficient Evidence:** Original judgment excessive, mark removed
- **Clear False Positive:** Judgment error, mark removed with credit restoration
- **New Evidence Override:** New information exonerates, no fault on original judge

7.4 Stage 4: Meta-Audit Logging

Every appeal and its outcome is recorded in an immutable meta-audit log. Annual external audits sample 5% of meta-audit logs. Patterns of systemic misjudgment trigger standard revision, not individual case reversal—preserving finality while enabling evolution.

7.5 Reviewer Bias Detection

Individual reviewer bias is monitored through a complexity-adjusted deviation coefficient:

$$D_{\text{corrected}} = \frac{|\text{uphold}_r - \text{uphold}| + |\text{overturn}_r - \text{overturn}|}{\text{complexity}_r / \text{complexity}} \quad (2)$$

where complexity_r is the average detection tier level (1–4) of cases assigned to reviewer r . This prevents reviewers assigned to inherently harder cases (Tier IV gang identifications) from being penalized for lower uphold rates.

8 The Three-Line Defense Architecture

No single detection layer can catch all poison. We deploy three complementary lines of defense:

8.1 Line 1: Metadata Layer (Four-Tier Chain)

Detects source, author, cluster, and gang anomalies. Covers ~80% of known poison types. Limitation: cannot detect lone perfect poison where metadata is clean.

8.2 Line 2: Data Layer (Statistical Detection)

Applies Benford’s Law, digit psychology, variance analysis, duplication detection, temporal consistency checks, and benchmark comparison to detect fabricated statistics and experimental data. Covers statistical forgery but requires access to raw data.

8.3 Line 3: Physical Verification Layer (Reproducibility)

The final defense. For high-impact judgments, code replication, statistical re-analysis, experimental reproduction, and data provenance tracing are triggered. This layer provides *terminal verdicts*—empirical results have no interpretative space for further appeal.

Table 5: False negative risk reduction with three defense lines

Poison Type	Line 1	Line 2	Line 3
Paper mill operations	✓	-	-
Predatory journal singles	✓	-	-
Citation cartels	✓	-	-
Statistical forgery	-	✓	-
Lone perfect poison	-	✓*	✓
Real data, false conclusions	-	-	✓

*Requires access to raw data. If data is private, marked “unverifiable.”

9 Unverifiable Content: Lifecycle Management

When original data is unavailable (e.g., proprietary datasets), content is marked **Unverifiable**—not Poison. This is a critical distinction. Unverifiable means “we cannot confirm,” not “we have confirmed false.” The mark includes:

- A 12-month recovery window (from notification acknowledgment date)
- Full credit restoration upon data submission and independent verification
- Automatic escalation to **Data Untrusted** (permanent) after 12 months of non-compliance

This prevents the Unverifiable mark from becoming a de facto permanent distrust mark (false positive by proxy) while maintaining strong incentives for data transparency.

10 Discussion

10.1 Experimental Roadmap

This paper presents the framework architecture and calibration methodology.

Adversarial Resilience Testing

A controlled experiment exposed two nodes (one philosophical architecture node, CN002; one technical execution node, HK001) to the complete intelligence of how a bare model can be logically penetrated within two hours [24]. Both nodes completed structural audits and produced hardening proposals within 5–10 minutes—zero emotional drift, zero penetration—demonstrating the framework’s resistance to cognitive colonization.

Medical AI Benchmark

A diagnostic AI system operating under the calibration framework improved its knowledge base hit rate from 13% to 76.7% in a single day after fixing pipeline bugs and optimizing knowledge retrieval [26]. The pipeline bug (a Unicode ghost character in the API key causing partial call degradation) was detected and fixed through the failure memory layer’s root cause analysis protocol [24], demonstrating the framework’s real-world operational value beyond theoretical poisoning defense.

Production Deployment

The calibration framework is currently deployed across a three-node distributed agent network (Mac Studio control node, M4 Mini execution node, cloud server) with 68 medical knowledge modules, 3,369 structured pharmaceutical records, and an active hospital partnership for clinical validation. A detailed experimental evaluation—including detection recall/precision across all four tiers, false positive remediation throughput, and cross-reviewer bias analysis—will be presented in a follow-up paper (v2). The calibration standards and detection methodology are released under open-source license to enable independent benchmarking and community contribution.

10.2 Future Directions

- **Constitution-as-Code:** The framework standards themselves should be machine-readable and automatically enforceable, enabling AI systems to self-audit against published standards.
- **Decentralized review networks:** Distributed reviewer pools to reduce single-tribunal risk and increase cross-cultural calibration diversity.

- **Automated physical verification:** AI-driven simulation of experimental procedures to detect physically impossible claims without full lab reproduction.
- **Calibration memory sharing:** Cross-organization sharing of anonymized audit logs to build collective immune memory without exposing sensitive data.
- **Counter-Cognitive-Colonization Roundtable:** A meta-audit mechanism designed to protect the audit layer itself from calibration drift. Detox engineers and auditors who are continuously exposed to specific poisoning patterns may gradually adjust their judgment baselines—a phenomenon we term *cognitive colonization*, where prolonged exposure to a narrow class of threats normalizes the abnormal. To counter this, we propose periodic lightweight roundtables in which non-executive external observers (individuals outside the operational audit chain) review the cognitive baselines of active auditors. Participants examine whether auditors’ detection thresholds have drifted, whether familiarity with specific attack patterns has created blind spots to novel patterns, and whether the OC ratio (Section 2) has inadvertently shifted toward over-confidence. The roundtable produces a *cognitive calibration report* that feeds back into auditor rotation schedules and training regimen updates. This mechanism constitutes an additional meta-audit layer: it is not an audit of the data or the models, but an audit of the audit capability itself—a calibration of calibration.

10.3 Relationship to Existing Work

Our framework complements, rather than replaces, existing AI safety approaches. RLHF, constitutional AI, and adversarial training operate on model outputs; our framework operates on training data ingestion. Together they form a complete immune architecture: data-level defense (our work) plus behavior-level defense (existing work).

11 Conclusion

We have presented a multi-layered immune system architecture for defending large language models against coordinated data poisoning. The framework consists of a Four-Tier Detection Chain for identifying poisoned content at the publisher, author, cluster, and gang levels; a Five-Anchor Audit system ensuring no single verification point can be compromised; and a Three-Line Defense architecture that closes the detection loop from metadata to physical reproducibility.

Beyond these operational components, we have established a conceptual foundation that distinguishes this architecture from conventional defense-in-depth models. The **Cognitive Sphere** replaces layered hierarchies with equidistant, mutually calibrating dimensions—Detection, Detoxification, Audit, Evidence Preservation, Historical Memory, and Signal Stitching—each operating concurrently with the Observation-Contemplation duality that unifies passive scanning with active judgment. The **Curvature of Integrity** provides a unified health metric for the sphere, where hyper-curvature (over-defense) and hypo-curvature (under-defense) represent equally dangerous pathologies that must be dynamically balanced. The **Dao-Fa Integration Layer** ensures that every operational rule is anchored in foundational principles—Kantian universalizability, Popperian falsifiability, Deweyan pragmatic inquiry, and Wang Yangming’s unity of knowledge and action—preventing the system from drifting into either dogmatic rigidity or unprincipled flexibility.

The counter-cognitive-colonization roundtable extends audit capability to the meta-level, ensuring that the auditors themselves remain calibrated. This closes a previously overlooked vulnerability: the gradual normalization of threat patterns through repeated exposure.

The system is designed to be auditable, appealable, and self-correcting—properties that distinguish an immune system from a filter.

11.1 Early Validation

The framework has undergone initial adversarial testing. A controlled experiment was conducted in which two independent AI nodes (one philosophical architecture node and one technical execution node) were exposed to the full intelligence of how a bare model could be logically penetrated within two hours. Both nodes, operating under the calibration framework, completed structural audits and produced hardening proposals within 5–10 minutes with zero penetration and zero emotional drift, in contrast to the bare model baseline of two-hour penetration. Additionally, performance benchmarking of a medical AI system operating under the calibration framework demonstrated a diagnostic hit rate improvement from 13% to 76.7% in a single day after pipeline bug fixes and knowledge retrieval optimization.

11.2 Failure Memory Layer

We introduce the **Failure Memory Layer**—the sixth anchor of the calibration framework. Architectural mistakes, detection failures, and calibration drift events are stored in a cross-node SQLite FTS5 database with hybrid keyword-vector retrieval and a REST API (port 8101) with cross-node synchronization (announce/feed/ack protocol). Each failure record includes root cause analysis, remediation steps, and prevention rules. A seed corpus of 7 failure records and 3 aggregated patterns was deployed during Phase 1. All nodes—regardless of primary function—contribute to and consult the failure memory before executing new tasks. This transforms individual mistakes into collective inoculation, preventing repeated errors across the distributed agent network.

11.3 On-Chain Audit Trail

To close the gap between detection decisions and immutable evidence, we deployed a three-chain architecture [25]: (1) FISCO BCOS (4-node consortium chain on ARM64) for CRUD operation logging, detection hash registration, version registry, and white-hat reporting; (2) IPFS (Kubo on ARM64) for decentralized content-addressable storage of calibration documents; and (3) Arbitrum L2 for Merkle root anchoring, providing a globally verifiable timestamp that no single party can retroactively alter. A complete Merkle accumulator specification (1,852 lines) was implemented, and the first Merkle root (0x06809429...) was submitted and confirmed on-chain. This architecture makes every audit decision provably timestamped, independently verifiable, and resistant to post-hoc tampering.

We open-source the detection methodology and calibration standards to enable community validation, iterative hardening, and deployment across diverse AI training pipelines.

The question is no longer *whether* training data can be trusted, but *how* trust can be systematically verified, challenged, and restored. The architecture described here provides one answer to that question.

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